

R&T

Cours Réseaux : HSRP

R&T



- Introduction
- Principes
- Configuration de base
- Configuration multigroup
- Conclusion

Introduction : HSRP

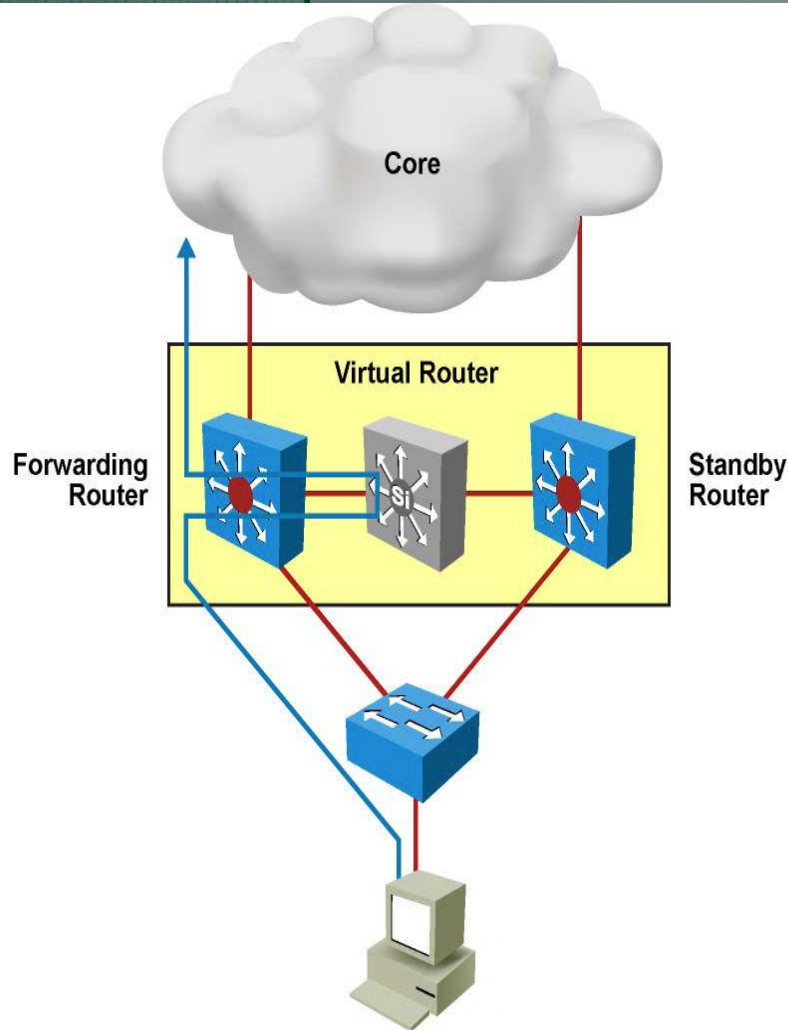
R&T



Hot Standby Router Protocol

- Propriétaire Cisco
- Système de redondance
- Haute disponibilité

Introduction (2)

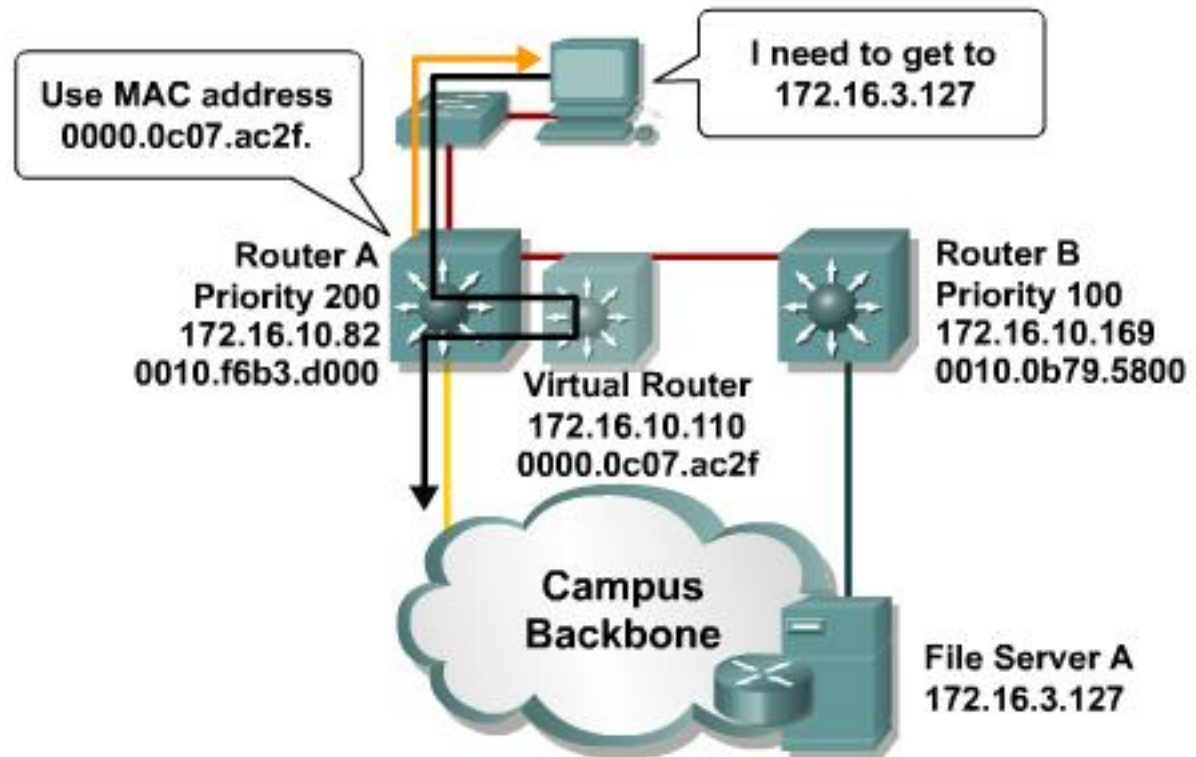


Hot Standby Router Protocol

- Routeur Actif
- Routeur en Standby
- Routeur Virtuel
- Messages Hello

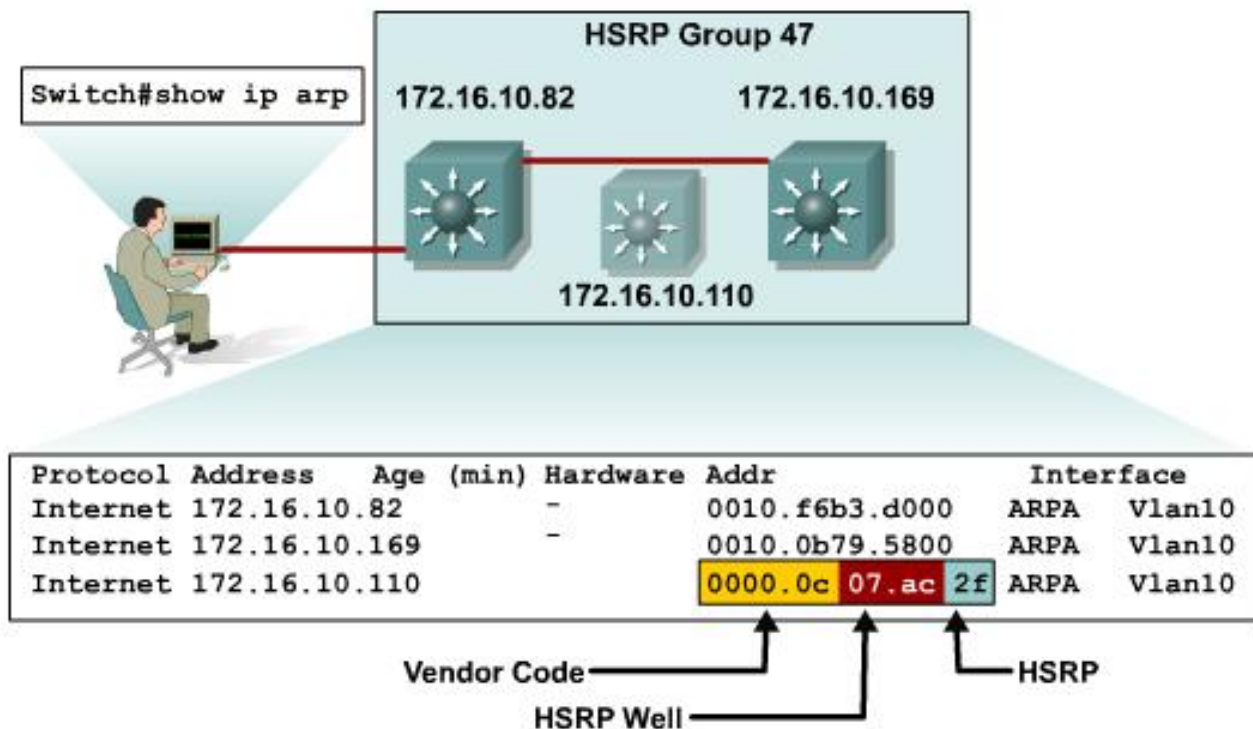
Routeur Actif

R&T

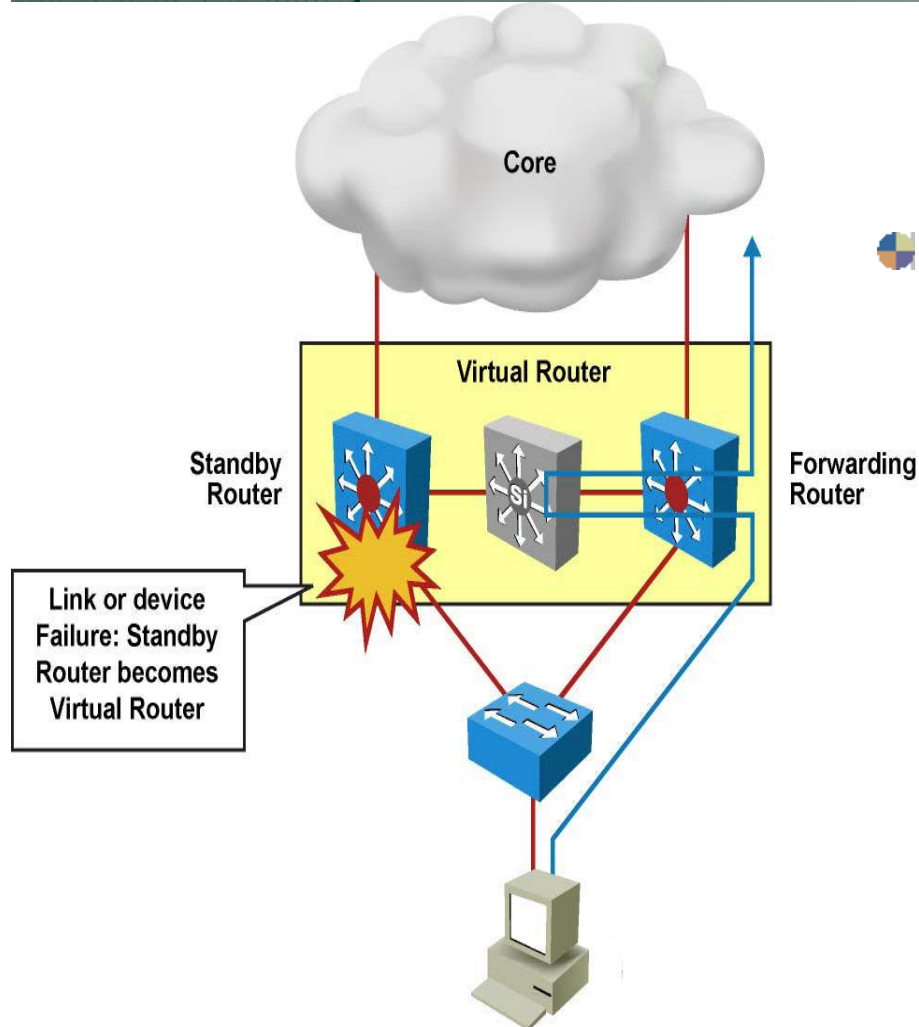


Routeur Virtuel

R&T



HSRP principes

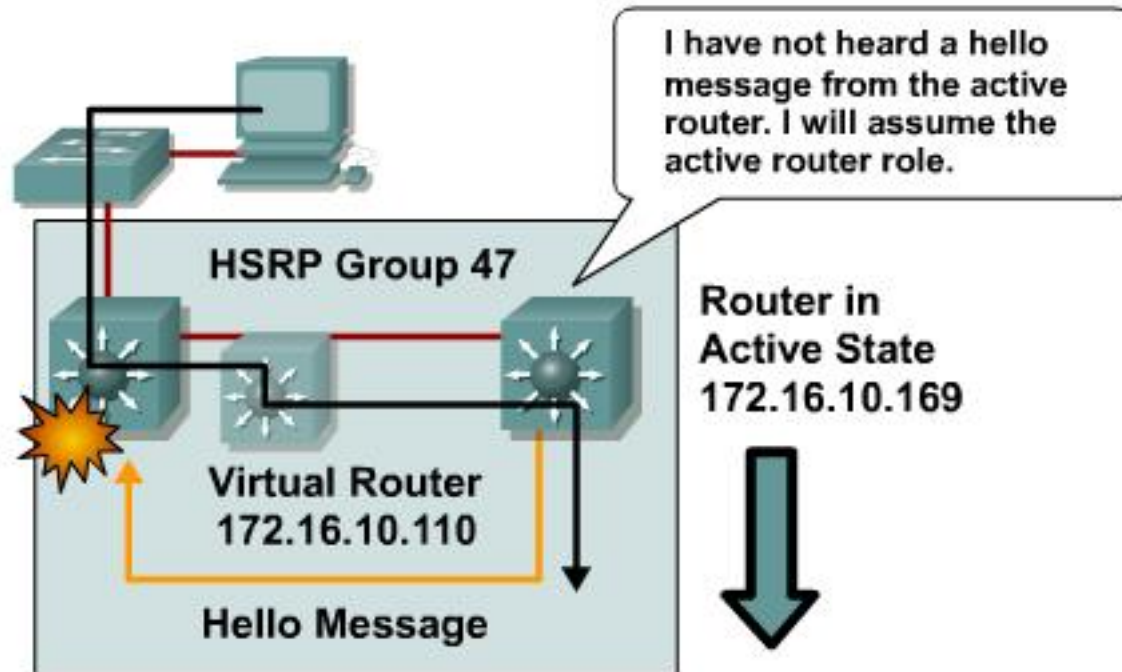


● Bascule en cas de panne (Fail over)

- Lien HS
- Routeur HS
- Message Hello non reçu
- Routeur en **Standby** devient le Routeur **Actif**
- Transparence pour PCs

Interactions

R&T



Etats de HSRP

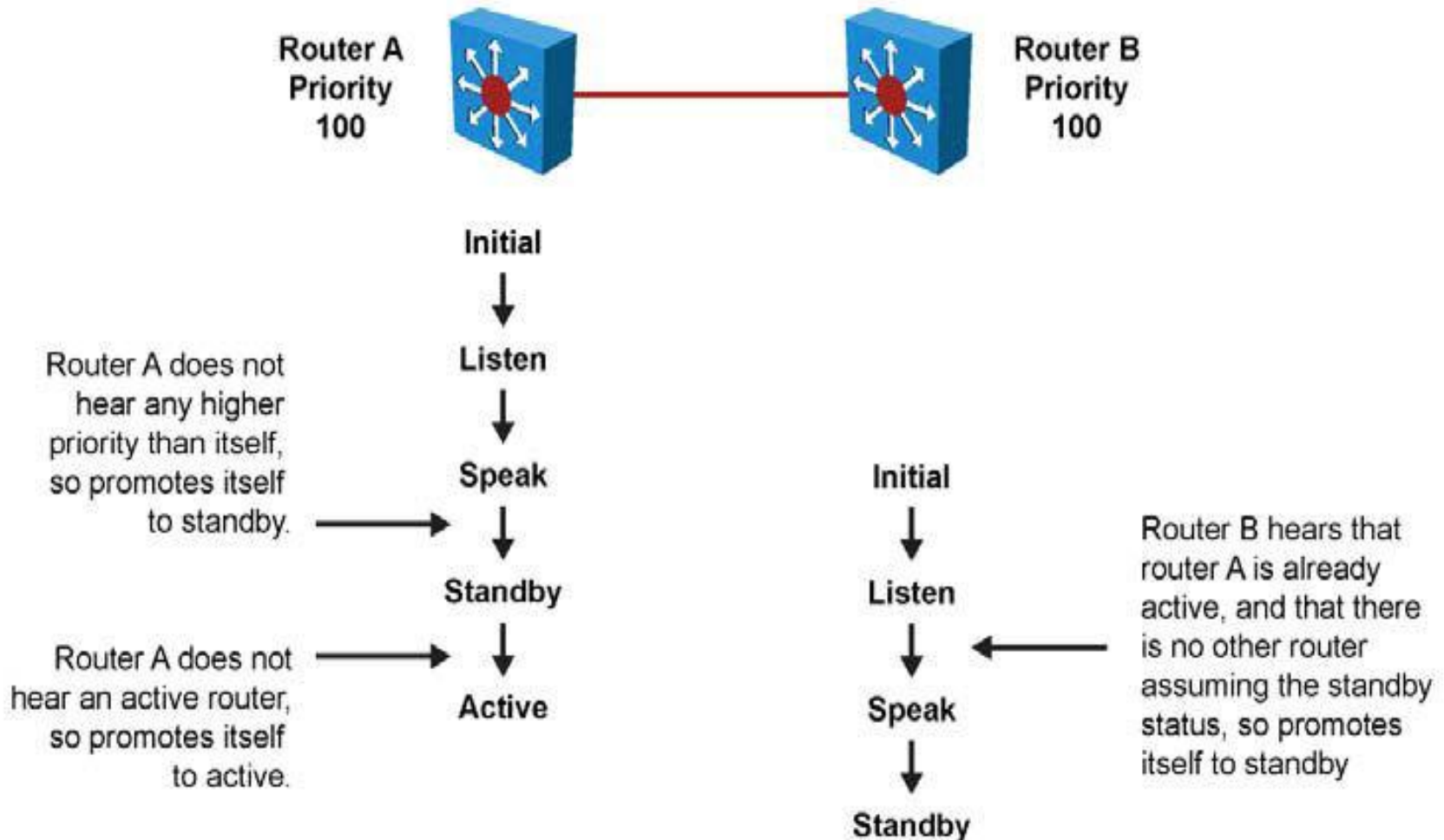


R&T

State	Definition
Initial	The beginning state. The initial state indicates that HSRP does not run. This state is entered via a configuration change or when an interface first comes up.
Listen	The router knows the virtual IP address, but the router is neither the active router nor the standby router. It listens for hello messages from those routers.
Speak	The router sends periodic hello messages and actively participates in the election of the active or standby router. A router cannot enter speak state unless the router has the virtual IP address.
Standby	The router is a candidate to become the next active router and sends periodic hello messages. With the exclusion of transient conditions, there is, at most, one router in the group in standby state.
Active	The router currently forwards packets that are sent to the group virtual MAC address. The router sends periodic hello messages. With the exclusion of transient conditions, there must be, at the most, one router in the active state in the group.

Transitions Etats HSRP

HSRP Standby Group 1



HSRP:Configuration



R&T

§ Router(config-if)#

§ *standby group-number ip ip-address*

HSRP:Configuration (2)

R&T



- group number : optional (0 par défaut)
- Priority : 0-255 par défaut 100
 - Si égalité, bascule vers prioritaire avec gp# la + élevée
- Preempt => respect des priorités

standby *group-number* **priority** *priority-value*

standby [*group-number*] **preempt** [**delay** {**minimum**
seconds **reload** **seconds** **sync** **seconds**}]

HSRP Configuration (3)

R&T

Exemple

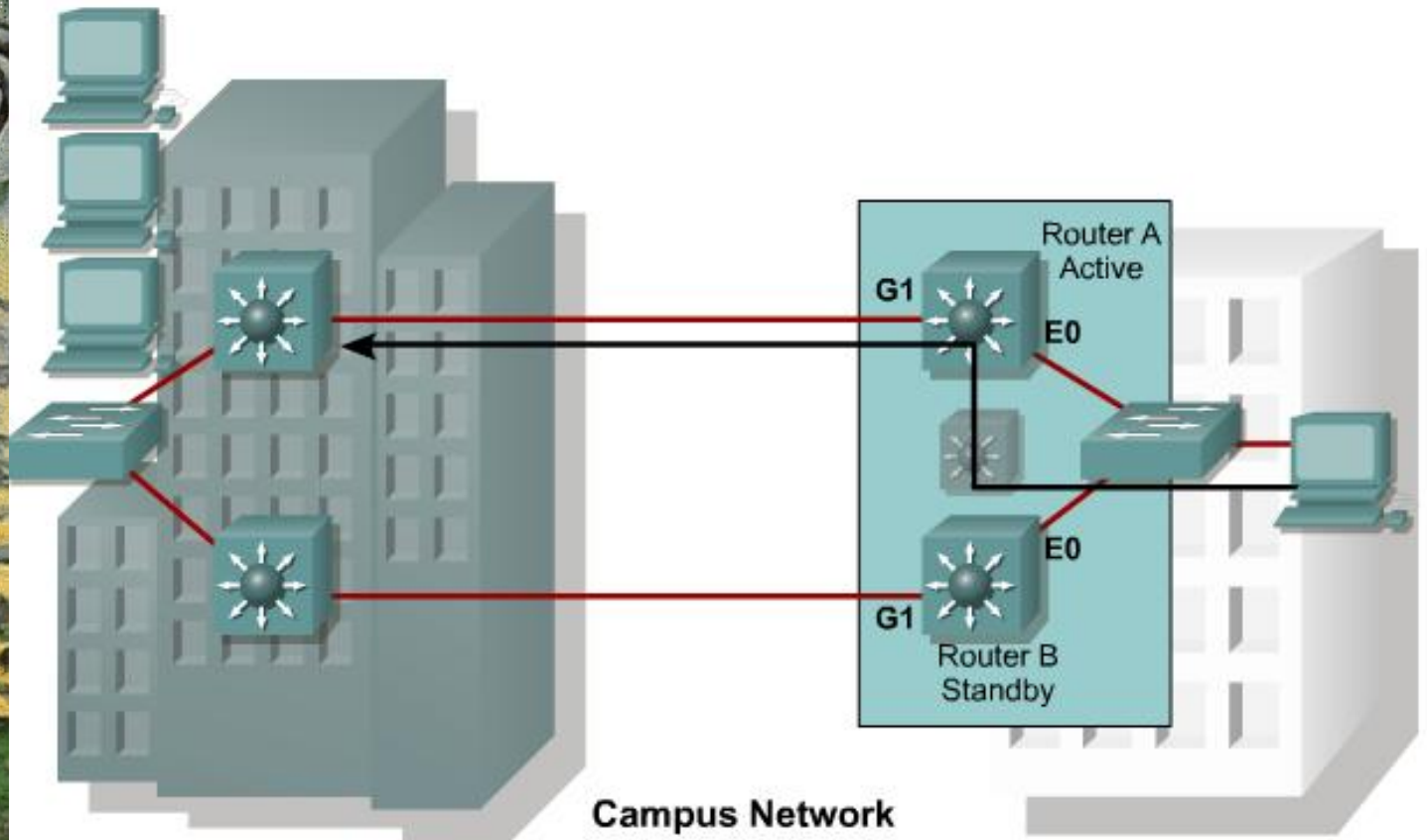
HSRP Standby Group 10



```
RouterA(config)#interface F0/10  
RouterA(config-if)#ip address 10.1.1.2 255.255.255.0  
RouterA(config-if)#standby 10 ip 10.1.1.1  
RouterA(config-if)#standby 10 priority 110  
RouterA(config-if)#standby 10 preempt
```

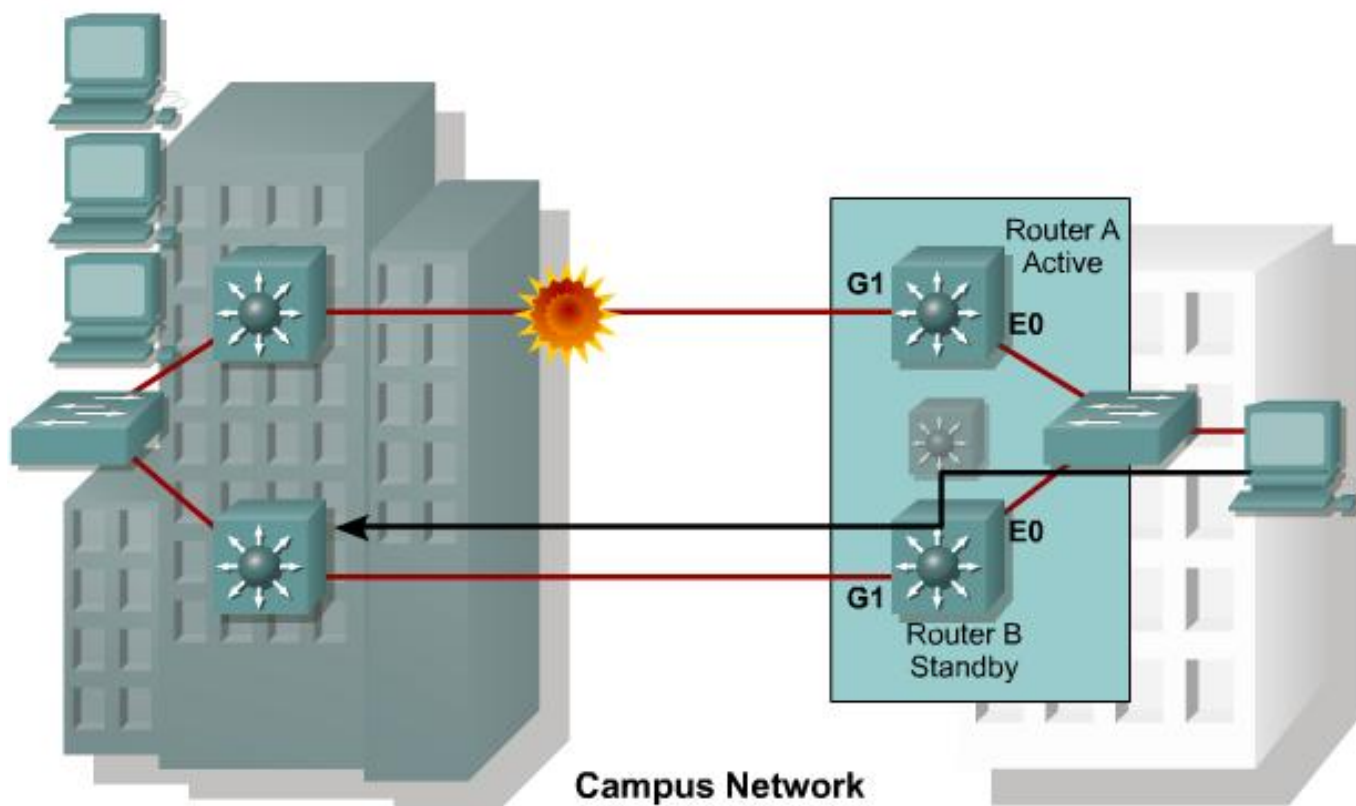

HSRP : Interface Tracking

R&T



HSRP : Interface Tracking(2)

R&T



HSRP : Interface Tracking(3)



R&T

```
Switch(config-if)#standby [group-number] track type number  
[interface-priority]
```

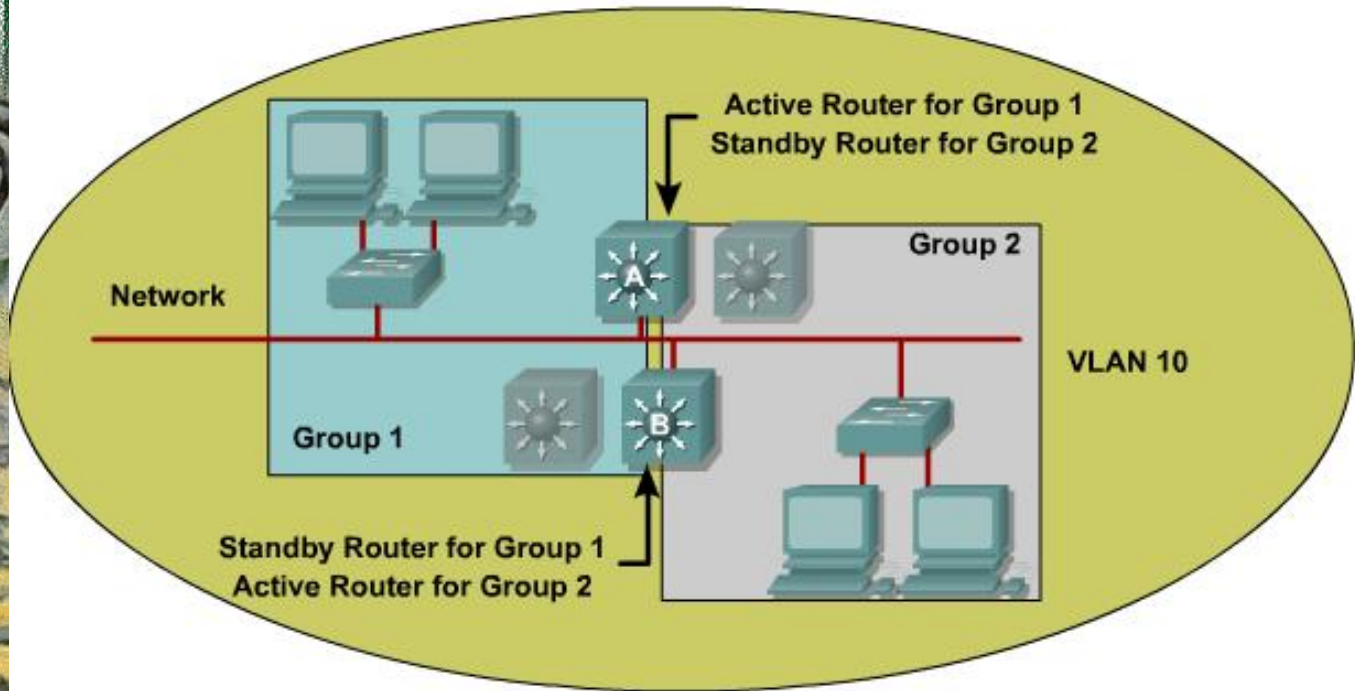
- Configures HSRP tracking

```
Switch(config)#interface vlan 10  
Switch(config-if)#standby 1 track GigabitEthernet 0/7 50  
Switch(config-if)#standby 1 track GigabitEthernet 0/8 60
```

- Example of HSRP tracking

HSRP : groupe multiple

R&T



■ Répartition de charge HSRP

- Affectation de groupes différents sur le même sous-réseau

Conclusion

R&T

- HSRP : outil de haute disponibilité
- Standby, priority, track
- Configuration multi-groupe
- VRRP / GLBP